

1. Introduction

Brand Name: Calpol® 650mg

Generic Name: Paracetamol Tablets IP 650 mg

Drug Class: Non-opioid analgesic and antipyretic

Calpol® 650 mg tablet is generally used for the symptomatic relief of mild to moderate pain and fever reduction, contains paracetamol 650 mg as the active ingredient. Despite being well-tolerated at the recommended dosage, some side effects, such as nausea, vomiting, and abdominal discomfort, are observed rarely that typically resolve with time. However, if these symptoms persist or worsen, seek immediate medical assistance.

2. Mechanism of Action

The mechanism by which paracetamol achieves its pharmacological effects is not fully understood but believed to involve the COX pathway inhibition in the CNS. This inhibition is effective in environments where the peroxide concentration is low, such as the brain. In contrast, in peripheral tissues and at sites of inflammation, where the peroxide concentration is high, the action of paracetamol on the COX is significantly reduced. While paracetamol is known for its analgesic and antipyretic properties, its anti-inflammatory action in the peripheral tissues is weak.

Additionally, paracetamol converts to a compound AM404, within the brain, which enhances the analgesic properties by interacting with certain receptors involved in pain modulation.

3. Pharmacokinetics

Oral paracetamol is completely absorbed from the gastrointestinal tract showing a peak plasma concentration within 30-60 minutes after ingestion. Plasma protein binding is minimal at usual therapeutic concentrations and diffuses easily through most body fluids and tissues. Primarily metabolised in the liver, paracetamol is

excreted through urine as conjugates of glucuronide and sulphate, with less than 5% as unchanged paracetamol. The mean plasma half-life is about 2.3 hours.

4. Indications and Therapeutic Efficacy

1. Mild to moderate pain

- Headache and acute migraine
- Muscle and joint pain
- Osteoarthritis
- Dental pain
- Management of postoperative pain.

2. Fever

3. Symptomatic relief of cold and flu- associated symptoms

5. Dosages and Administration

➤ Adults and adolescents (12 years and older)

Dose: 650 mg orally every 4 to 6 hours as needed

Maximum daily use: ≤ 4 g (4000 mg) in 24 hours

Warning

Do not exceed the recommended dose. In suspected overdose, immediate medical assistance is necessary.

➤ Specific Patient Populations

- *Hepatic impairment*: Paracetamol should be used with caution in patients diagnosed with hepatic illness.
- *Renal impairment*: Paracetamol should be used with caution in patients diagnosed with renal impairment (GFR < 60 mL/min). Following medical advice before taking the medication is recommended.
- *Pregnancy considerations*: When clinically indicated paracetamol may be used during pregnancy. Current data show no adverse pregnancy outcome when used as directed. The lowest effective dose and shortest duration of treatment should be considered.

- *Lactation considerations:* As the amount detected in breast milk is clinically insignificant at the recommended doses, paracetamol is not contraindicated in lactating mothers.
- *Elderly people:* No dosage adjustment is required, however caution in patients with chronic hepatic impairment, renal impairment, and excessive alcohol usage.
- *Children:* The safety and efficacy of Calpol® 650 mg tablets have not been established for children under 12 years of age.

➤ **Effects on Ability to Drive and Use Machines**

Does not affect the ability to drive or use machines.

6. Adverse Effects

Paracetamol exhibits good tolerability at the recommended dose. Potential adverse reactions include nausea, vomiting, diarrhoea, abdominal discomfort, and skin rashes. Overdose of paracetamol is associated with hepatic and renal impairment.

➤ **Overdose**

During the initial 24 hours, patients may be asymptomatic or may exhibit symptoms that include nausea, vomiting, pallor, and abdominal pain. Acute paracetamol overdosage may lead to hepatic injury after 24-48 hours, and potential worsening over a period of 4 to 6 days.

Treatment includes the administration of N-acetylcysteine, up to 24 hours of paracetamol ingestion, with maximum protective effect if given within 8 hours post ingestion.

7. Contraindications

Paracetamol is contraindicated in patients with

1. Prior hypersensitivity to paracetamol
2. Severe hepatic illness

8. Drug Interactions

Prolonged, regular oral administration of paracetamol in a patient using warfarin may increase the risk of bleeding. Dosage adjustment may be required.

Chronic alcohol use increases the risk of hepatic toxicity with paracetamol.

Combinations of traditional NSAID–paracetamol may provide greater analgesia than paracetamol alone, regardless of the route administered.

The concomitant use of any other products containing paracetamol may lead to overdose.

9. Storage and Handling

Store below 30°C. Protect from light and moisture. Keep out of reach of children.

10. Summary

Paracetamol, also known as Acetaminophen, is a non-opioid and antipyretic drug used for the management of pain and fever. It is a widely used analgesic in the world and serves as the first step in the WHO analgesic ladder for treating cancer - associated pain. It is not completely understood how exactly paracetamol works but thought to involve the inhibition of prostaglandin synthesis in the central nervous system. The anti-inflammatory action of paracetamol is weak compared to other non-steroidal anti-inflammatory drugs. Paracetamol is rapidly absorbed from the gastrointestinal tract following oral administration. The rate of absorption is dependent on gastric emptying that may be delayed by food, pethidine, diamorphine, and enhanced by metoclopramide. Paracetamol is well tolerated when taken with caution, but care should be taken to prevent an overdose that may lead to severe toxicity.

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